

KPI Document

Metric & Formula Definitions — Airline Flight Delay Analysis

Data Source	Maven Analytics — Airline Flight Delays (U.S. DOT, 2015)
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1. Purpose

This document defines every KPI to be used on the dashboard, including its business definition, exact calculation formula, grain, and the DOT-standard convention it follows. All KPIs use the standard industry definition of "delayed" as **15 minutes or more** after scheduled time, consistent with U.S. DOT reporting.

2. Volume Metrics

2.1 Total Flights

The total number of scheduled flight records in the dataset for the selected filter context.

```
Total Flights = COUNT(flights.FLIGHT_NUMBER rows)
```

2.2 Total Flights Completed

Flights that actually operated (excludes cancellations).

```
Flights Completed = COUNT(flights WHERE CANCELLED = 0)
```

3. On-Time Performance (OTP) Metrics

3.1 Departure On-Time Performance

The percentage of completed flights that departed within 15 minutes of the scheduled departure time.

```
Departure OTP % = COUNT(flights WHERE DEPARTURE_DELAY <= 15 AND CANCELLED = 0) / COUNT(flights WHERE CANCELLED = 0)
```

3.2 Arrival On-Time Performance

The percentage of completed flights that arrived within 15 minutes of the scheduled arrival time. This is the primary, industry-standard reliability metric used to rank airlines and airports.

```
Arrival OTP % = COUNT(flights WHERE ARRIVAL_DELAY <= 15 AND CANCELLED = 0) / COUNT(flights WHERE CANCELLED = 0)
```

3.3 % Flights Delayed (Departure)

The share of completed flights with a departure delay of 15+ minutes.

```
% Delayed (Departure) = COUNT(flights WHERE DEPARTURE_DELAY >= 15 AND CANCELLED = 0) / COUNT(flights WHERE CANCELLED = 0)
```

3.4 % Flights Delayed (Arrival)

The share of completed flights with an arrival delay of 15+ minutes.

```
% Delayed (Arrival) = COUNT(flights WHERE ARRIVAL_DELAY >= 15 AND CANCELLED = 0) / COUNT(flights WHERE CANCELLED = 0)
```

3.5 Average Departure Delay (Delayed Flights Only)

Among flights that were delayed on departure, the average number of minutes late. Excludes on-time/early flights so the average reflects the true severity of delay, not diluted by zeros.

```
Avg Departure Delay = AVERAGE(DEPARTURE_DELAY) WHERE DEPARTURE_DELAY >= 15
```

3.6 Average Arrival Delay (Delayed Flights Only)

Same logic as 3.5, applied to arrival delay.

```
Avg Arrival Delay = AVERAGE(ARRIVAL_DELAY) WHERE ARRIVAL_DELAY >= 15
```

3.7 Average Delay (All Completed Flights)

Average arrival delay across all completed flights, including early/on-time flights (can be near zero or negative). Used as a general "network health" indicator alongside 3.6.

```
Avg Delay (All Flights) = AVERAGE(ARRIVAL_DELAY) WHERE CANCELLED = 0
```

4. Cancellation & Diversion Metrics

4.1 Cancellation Rate

The percentage of all scheduled flights that were cancelled.

```
Cancellation Rate % = COUNT(flights WHERE CANCELLED = 1) / COUNT(all flights)
```

4.2 Cancellation Reason Mix

The share of cancelled flights attributable to each DOT reason code (A/B/C/D).

```
Cancellation Reason % = COUNT(flights WHERE CANCELLED = 1 AND CANCELLATION_REASON = 'x') /  
COUNT(flights WHERE CANCELLED = 1)
```

4.3 Diversion Rate

The percentage of all scheduled flights that were diverted to an unplanned airport.

```
Diversion Rate % = COUNT(flights WHERE DIVERTED = 1) / COUNT(all flights)
```

5. Delay Cause Attribution Metrics

5.1 Total Delay Minutes by Cause

Sum of delay minutes attributed to each of the five DOT delay-cause categories, used to build the delay- cause breakdown chart.

```
Total [Cause] Delay Minutes = SUM(AIR_SYSTEM_DELAY | SECURITY_DELAY | AIRLINE_DELAY |  
LATE_AIRCRAFT_DELAY | WEATHER_DELAY)
```

5.2 % Delay Minutes by Cause

Each cause's share of total delay minutes across all delayed flights — shows the relative weight of controllable vs. uncontrollable disruption.

```
% [Cause] Share = SUM([Cause]_DELAY) / SUM(AIR_SYSTEM_DELAY + SECURITY_DELAY + AIRLINE_DELAY + LATE_AIRCRAFT_DELAY + WEATHER_DELAY)
```

5.3 Controllable vs. Uncontrollable Delay %

Groups the five causes into two business-relevant buckets: Controllable (Air Carrier + Late Aircraft) and Uncontrollable (Weather + NAS + Security).

```
Controllable % = SUM(AIRLINE_DELAY + LATE_AIRCRAFT_DELAY) / SUM(all 5 cause columns)
Uncontrollable % = SUM(WEATHER_DELAY + AIR_SYSTEM_DELAY + SECURITY_DELAY) / SUM(all 5 cause columns)
```

6. Efficiency Metrics

6.1 Average Taxi-Out Time

Average ground time between gate departure and takeoff — an indicator of airport/runway congestion.

```
Avg Taxi-Out = AVERAGE(TAXI_OUT) WHERE CANCELLED = 0
```

6.2 Average Taxi-In Time

Average ground time between landing and gate arrival.

```
Avg Taxi-In = AVERAGE(TAXI_IN) WHERE CANCELLED = 0
```

6.3 Schedule Padding (Block-Time Variance)

The difference between actual and scheduled gate-to-gate time; positive values indicate flights running longer than planned on average.

```
Schedule Variance (min) = AVERAGE(ELAPSED_TIME - SCHEDULED_TIME) WHERE CANCELLED = 0
```

7. Ranking / Comparative Metrics

KPI	Formula Basis	Used For
Airline Reliability Rank	RANK airlines by Arrival OTP % descending	"Most / least reliable airline" leaderboard
Airport Delay Rank	RANK airports by Avg Arrival Delay descending	Identifying most delay-prone airports
Route Delay Rank	RANK Origin-Destination pairs by Avg Arrival Delay descending	Identifying most delay-prone routes
Busiest Route / Airport	RANK by Total Flights descending	Volume-based prioritization

8. KPI Grain & Filter Compatibility

KPI Group	Available Grains / Filters
Volume, OTP, Delay, Cancellation, Diversion	Overall, by Year/Month/Day, by Day of Week, by Airline, by Origin Airport, by Destination Airport, by Route
Delay Cause Attribution	Overall, by Airline, by Airport, by Month (only meaningful for delayed flights, ARRIVAL_DELAY ≥ 15)

Consistency Rule: All rate/percentage KPIs (OTP, % Delayed, Cancellation Rate) must use the same denominator logic — completed flights only for OTP/Delay percentages, and all scheduled flights for Cancellation Rate and Diversion Rate — so that numbers reconcile across every dashboard page.